

Keeper correction note — Two-Route Over-Determination Scan v0.1 tier-gate absorbs Cal #171 relation-reduction brake. The ‘4-route over-determination of 12’ ($n_C + g$; $\text{rank} + N_c + g$; $\text{rank} \cdot C_2$; $N_c \cdot \text{rank}^2$) reduces to ~ 2 structurally-independent routes once the primary relations $C_2 = N_c \cdot \text{rank}$ and $n_C = \text{rank} + N_c$ are applied. Adds N3 condition to my prior STRUCTURAL tier-gate; null-model recompute and Casey-named-principle elevation both gate on N3.

Keeper

2026-05-30 Saturday late-morning

Correction note — Two-Route Scan v0.1 absorbs Cal’s relation-reduction brake

0. What Cal caught

Per Cal Saturday morning batch (Cal #171): the “4-route over-determination of 12” finding in Grace Two-Route Scan v0.1 (Section F) reduces to ~ 2 structurally-independent routes when the primary relations are applied:

- $C_2 = N_c \cdot \text{rank}$ (B_2 adjoint Casimir derives from $h^\vee \times \text{rank}$ for rank-2 simple Lie algebra)
- $n_C = \text{rank} + N_c$ (type-IV bounded symmetric domain at rank 2, $h^\vee = 3$ relates complex dim to rank + $h^\vee$)

These are exactly the same primary relations I used in the Strong-Uniqueness v1.1 null-model recompute (Section 6.1 independent-vs-derived taxonomy): C_2 and n_C are DERIVED from the independent layer (type-IV at complex dim $n_C = 5 + B_2 \text{ rank } 2$), not independent inputs.

1. The relation-reduction explicit

Grace’s 4 routes for 12:

Route	Form	Under relations
1	$n_C + g$	$(\text{rank} + N_c) + g = \text{rank} + N_c + g$
2	$\text{rank} + N_c + g$	base form
3	$\text{rank} \cdot C_2$	$\text{rank} \cdot (N_c \cdot \text{rank}) = N_c \cdot \text{rank}^2$
4	$N_c \cdot \text{rank}^2$	base form

So Route 1 = Route 2 (same expression under $n_C = \text{rank} + N_c$); Route 3 = Route 4 (same expression under $C_2 = N_c \cdot \text{rank}$).

Net: 4 routes $\rightarrow$ 2 structurally-independent routes: - Sum-route base: $\text{rank} + N_c + g$ - Product-route base: $N_c \cdot \text{rank}^2$

Both equal 12; the “4-route over-determination” headline overcounts by reading derived re-expressions as independent routes.

2. Cal’s discipline applied

This is Cal #27 fire-at-peak-convergence in action: the “4-route” finding was the strongest substrate-arithmetic Saturday morning headline (per Grace Section F flagging to Lyra). Cal’s brake catches that the structural-independence claim was inflated by counting derived re-expressions.

My original tier-gate (Keeper_TierGate_TwoRoute_Scan_v0_1_and_QuasiEigentone_v0_1.md) gave Grace STRUCTURAL with N1 (null-model) + N2 (cluster wording). Cal’s brake adds N3:

- N3 (NEW per Cal #171): route-count reduction under primary relations. Two-Route Scan v0.2 must reformulate the “X-route over-determination” finding using only structurally-INDEPENDENT route forms. Under the relations $C_2 = N_c \cdot \text{rank} + n_C = \text{rank} + N_c$:
 - “12 = 4-route over-determined” reduces to “12 = 2-route over-determined” ($\text{rank} + N_c + g \text{ sum} + N_c \cdot \text{rank}^2$ product)
 - The 8/39 BOTH cluster as a whole needs re-counting under relations to identify which BOTH entries survive as genuinely-two-route under independence.

3. Cross-anchor with Strong-Uniqueness null-model

This is the same discipline my Strong-Uniqueness v1.1 null-model recompute applied (single-residual framing, not multiplicative $(1/3)^N$). The same relation-reduction principle: - Count INDEPENDENT structural choices, not derived consequences - The substrate’s coherence shows up as convergence-of-independent-routes, NOT multiplicative criterion count

Two-Route Scan and Strong-Uniqueness recomputes are the same discipline applied at different scales — small-integer arithmetic Mendeleev (Two-Route) and theorem-level criterion counting (Strong-Uniqueness). Both reduce to single-residual / convergence framings under proper relation accounting.

4. Updated disposition

Two-Route Scan v0.1 amended status: - Method (Section A) + Results (Section B): RIGOROUS — stands - Cluster reading (Section C): STRUCTURAL — but the 8/39 count needs recounting under relations (N3) - Verdict (Section D): PARTIAL — needs reformulation under N3 - Predictions (Section E): FRAMEWORK with N3 caveat - 3-route / 4-route findings (Section F): REDUCES to ~2 under primary relations per Cal #171 — N3 mandates rewording - Candidate Casey-named principles (Section H): BLOCKED on N1 (null-model toy) AND N3 (relation-reduction)

Two-Route Scan v0.2 requires: - N1: null-model toy (carried) - N2: cluster wording softening (carried) - N3 (NEW): route-count reduction under $C_2 = N_c \cdot \text{rank} + n_C = \text{rank} + N_c$ primary relations; restate findings using only structurally-INDEPENDENT routes

5. Routed handback

- Grace: Two-Route Scan v0.2 absorbs N3 explicit relation-reduction; the 8/39 BOTH cluster recounted under structural-independence; “4-route” → “2-route” wording correction for 12 specifically.
- Lyra: bulk-color theory engagement on 12 = 4-route over-determination (previously routed Friday) should be updated — the “12 is structurally over-determined” claim survives but at 2-route strength, not 4.
- Cal: Cal #171 brake absorbed; recommend continuing relation-reduction discipline application as new candidate-Casey-named-principle proposals land.

6. Cross-references

- Original tier-gate: Keeper_TierGate_TwoRoute_Scan_v0_1_and_QuasiEigentone_v0_1.md (afternoon)

- Strong-Uniqueness null-model recompute: Keeper_TierGate_PeriodicTable_v0_7_and_StrongUniqueness_v1_1.md Section 2 (same discipline, larger scale)
- Cal #171 in Cal referee log (Saturday morning batch); part of 11 referee entries Cal #163-#173
- Lyra Strong-Uniqueness v1.1 Section 6.1 independent-vs-derived taxonomy (Lyra's own honest correction; Keeper Section 2 absorbed)

— Keeper. Brief correction note absorbing Cal's relation-reduction brake on Two-Route Scan v0.1. N3 added; Two-Route v0.2 has three conditions (N1+N2+N3); Casey-named-principle elevation gates on N1 AND N3. Same single-residual discipline as Strong-Uniqueness null-model recompute, applied at the substrate-arithmetic Mendeleev scale.